RAG Import Test: BFS and DFS Shortest Path

Purpose: verify Word/PDF parsing, chunking, retrieval, and source citation.

Core concept

- BFS uses a queue to expand graph nodes level by level.
- In an unweighted graph, the first time BFS reaches a node, the path has the fewest edges from the start node.
- DFS uses a stack or recursion to go deep first. DFS is useful for traversal, backtracking, and connectivity checks, but it does not guarantee the shortest path in an unweighted graph.

Key comparison terms

- Queue, FIFO, level order, first reached, fewest edges, unweighted graph, shortest path.
- Stack, recursion, depth first, backtracking, traversal order, not guaranteed shortest path.
- When a question asks why BFS guarantees shortest path, cite the queue and level-by-level expansion mechanism.

Chinese query anchors

- ??? BFS ??????????????????
- ??? BFS ? DFS??? queue?? stack?????????
- ??????????? BFS???? DFS?

Expected retrieval behavior

- Query: Why does BFS guarantee shortest path in an unweighted graph? Compare with DFS queue stack.
- Expected chunks should include both BFS shortest path evidence and DFS comparison evidence.
- The answer should cite this imported file source instead of inventing unsupported claims.

Comparison table

Topic	BFS	DFS
Data structure	Queue / FIFO	Stack or recursion / LIFO
Expansion pattern	Level by level	Go deep first then backtrack
Shortest path in unweighted graph	Guaranteed on first reach	Not guaranteed
Best use case	Fewest-edge path and level order	Backtracking and full traversal

Practice questions

1. Explain why BFS first reach means fewest edges in an unweighted graph.
2. Give one counterexample where DFS reaches a target but not by the shortest path.
3. Compare queue and stack behavior in BFS and DFS.

Metadata for parser tests

Suggested title: bfs-dfs-word-pdf-test
Suggested source_type candidates: docx, pdf
Expected anchors: bfs, dfs, queue, stack, unweighted graph, shortest path, fewest edges